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Name Brandon Buchanan Quiz 1 1/16/2007 Math for CS

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, \{1\}, \{\{1\}\}\}$

1. Is $\emptyset \subseteq A$? *yes*
2. Is $\emptyset \in A$? *no*
3. Is $\{1\} \in A$? *no*
4. Is $\{1\} \subseteq A$? *yes*
5. Is $2 \notin A$? *no*
6. What is $|A|$? *4*
7. Is $A \subset \{1, 2, 3, 5\}$? *yes*
8. Is $A \in \{1, 2, 3, 4, 5\}$? *no*
9. Is $\{1\} \in B$? *yes*
10. Is $\{1\} \subseteq B$? *yes*
11. Is $\{\{1\}\} \in B$? *yes*
12. Is $\{\{1\}\} \subseteq B$? *yes*
13. Is $\{\{\{1\}\}\} \in B$? *no*
14. Is $\{\{\{1\}\}\} \subseteq B$? *yes*
15. What is $P(A)$? *$\{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$*
16. What is $|P(A)|$? *16*